

# VANESSA CARLING KWONG

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**RESEARCH INTERESTS** Two-dimensional (2D) semiconducting materials have been of prominence in electronics development attributed to their excellent features for creating tunable devices from an easily isolable bulk state, mechanically cleaved at interlayer van der Waals (vdW) forces. I am interested in leveraging these materials, specifically transition metal dichalcogenides (TMDCs), and their unique properties to explore cutting-edge optical (e.g., correlated states) and magnetotransport (e.g., Quantum Hall Effect) phenomena for their implementation in advancing optoelectronics development.

**PROFESSIONAL INTERESTS** Interdisciplinary research consolidated with involvement in undergraduate research advocacy has deepened my commitment to teaching and effectively communicating science to all audiences. My interest lies in continuing research advocacy at the foundational level, to the extension of protecting these vital research works.

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**EDUCATION** **University of California, Riverside** Riverside, CA  
*Doctor of Philosophy, Physics (program coursework completed)* 2025 - 2026  
• Advisor: Prof. Andrew Joe

*Bachelor of Science, Physics* 2020 - 2025  
• Senior Thesis Advisor: Prof. Andrew Joe

**RESEARCH EXPERIENCE** **The Joe Lab at UC Riverside** Riverside, CA  
*Graduate Student Researcher* 2025.09 - Present

*External Researcher* 2025.04 - 2025.08  
• Engineer Ohmic contacts to p-type monolayer WSe<sub>2</sub> and MoSe<sub>2</sub>, measuring quantum magnetotransport  
• Fabricate devices and study electrical tuning of triplet exciton emission through optical characterisation  
• Enhance technical nanofabrication skills through training and assisting students; increase familiarity with cross-campus facilities at UC Irvine - Irvine Materials Research Institute

*Undergraduate Researcher* 2023.10 - 2025.04  
• Develop expertise in fabricating reliable vdW devices through precise patterning and dry transfer of mechanically exfoliated 2D materials  
• Perform and analyze optical and transport measurements through MATLAB programming  
• Set up new laboratory space and improve lab efficiency by building equipment and training incoming lab members on technical methods

*Research in Science & Engineering (RISE) Student Researcher* 2024.06 - 2024.08  
• Advisor: Prof. Andrew Joe  
• Develop low-temperature Pt contacts to monolayer TMDs; Test contact resistance and WSe<sub>2</sub> quality  
• Assist in fabricating graphene devices in search of the Quantum anomalous Hall Effect

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| TEACHING<br>EXPERIENCE     | UC Riverside, Department of Physics & Astronomy                                                                                                                                                  | Riverside, CA     |
|                            | <i>Teaching Assistant I</i><br><ul style="list-style-type: none"> <li>Facilitate weekly laboratory sections (PHYS 02LB, 02LC) and develop quiz material</li> </ul>                               | 2025.09 - Present |
| PROFESSIONAL<br>EXPERIENCE | Skyworks Solutions, Inc.                                                                                                                                                                         | Thousand Oaks, CA |
|                            | <i>Process Engineer I</i><br><ul style="list-style-type: none"> <li>Photolithography and etch oriented</li> </ul>                                                                                | 2026.07 - Present |
| PROJECTS                   | Triplet exciton light emission in MoSe <sub>2</sub> /WSe <sub>2</sub> Heterobilayer Devices                                                                                                      |                   |
|                            | <i>The Joe Lab, UC Riverside Dept. of Physics &amp; Astronomy</i>                                                                                                                                | 2025.07 - Present |
|                            | Magnetotransport in Monolayer TMD Heterostructure Devices                                                                                                                                        |                   |
|                            | <i>The Joe Lab, UC Riverside Dept. of Physics &amp; Astronomy</i>                                                                                                                                | 2024.06 - Present |
|                            | Pulsed Nuclear Magnetic Resonance                                                                                                                                                                |                   |
|                            | <i>PHYS 142W, UC Riverside</i>                                                                                                                                                                   | 2024.11 - 2024.12 |
|                            | Gamma Ray Emission of Radioactive Isobars                                                                                                                                                        |                   |
|                            | <i>PHYS 142W, UC Riverside</i>                                                                                                                                                                   | 2024.11           |
| PRESENTATIONS              | Noise Fundamentals                                                                                                                                                                               |                   |
|                            | <i>PHYS 142W, UC Riverside</i>                                                                                                                                                                   | 2024.10           |
|                            | MATLAB Laser Dispersion Program                                                                                                                                                                  |                   |
|                            | <i>The Joe Lab, UC Riverside Dept. of Physics &amp; Astronomy</i>                                                                                                                                | 2024.03 - 2024.04 |
|                            | Mutagenesis of KOIN and WFL in <i>A. thaliana</i>                                                                                                                                                |                   |
|                            | <i>Dynamic Genome, UC Riverside</i>                                                                                                                                                              | 2021.02 - 2021.03 |
|                            | [8] "Transport in Monolayer WSe <sub>2</sub> Devices,"                                                                                                                                           |                   |
|                            | <i>UC Riverside Dept. of Physics and Astronomy Senior Research Symposium, Riverside CA (2025.06)</i>                                                                                             |                   |
|                            | [7] "Research and Advances in Transport Within 2D Semiconducting Materials",                                                                                                                     |                   |
|                            | <i>UC Riverside CNAS Undergraduate Student Research Presentations for Rep. Mark Takano, Riverside CA (2025.04)</i>                                                                               |                   |
|                            | [6] Undergraduate Research Panelist,                                                                                                                                                             |                   |
|                            | <i>California State University, San Bernardino, calSWIFT'25 (California Semiconductor Workforce Development &amp; Innovation for Future Technologies) Symposium, San Bernardino CA (2025.04)</i> |                   |
|                            | [5] "Advances in Monolayer TMD Heterostructure Devices,"                                                                                                                                         |                   |
|                            | <i>UC Riverside PHYS 142W Capstone Presentation, virtual presentation (2025.03)</i>                                                                                                              |                   |
|                            | [4] "Quantum Transport in Monolayer TMD Heterostructure Devices,"                                                                                                                                |                   |
|                            | <i>UC Riverside Research in Science and Engineering (RISE) Symposium, Riverside CA (2024.08)</i>                                                                                                 |                   |
|                            | [3] "Optical Characterization of Two-Dimensional Semiconductor Heterostructures,"                                                                                                                |                   |
|                            | <i>UC Riverside Undergraduate Research &amp; Creative Activities Symposium, virtual presentation (2024.05)</i>                                                                                   |                   |
|                            | [2] "Mutating AT3G08680 to Determine Functional Redundancy in <i>Arapidopsis</i> ,"                                                                                                              |                   |
|                            | <i>UC Riverside Dynamic Genome, virtual presentation (2021.03)</i>                                                                                                                               |                   |
|                            | [1] "Love Gloves?"                                                                                                                                                                               |                   |
|                            | <i>Los Angeles County Science and Engineering Fair, Pasadena CA (2018.04)</i>                                                                                                                    |                   |

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| AWARDS                | • <b>R. Stephen White Scholarship</b> , UC Riverside Dept. of Physics & Astro.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025.05           |
|                       | • <b>Dean's List</b> (multiple), UC Riverside CNAS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2020 - 2025       |
|                       | • <b>Third Award in Microbiology Senior</b> , LACSEF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2018.04           |
| UNIVERSITY<br>SERVICE | <b>Science Olympiad Tournament - Grader</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                   |
|                       | <i>2026 Highlander Invitational - Experimental Design B, UC Riverside</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026.01           |
|                       | <b>Students Transforming Through Research (STR) - Undergraduate Advocate</b><br><i>Council on Undergraduate Research, Washington DC</i>                                                                                                                                                                                                                                                                                                                                                                                                                             | 2024.10 - 2025.03 |
| SKILLS                | <p><b>Laboratory:</b> Device fabrication (mechanical exfoliation, dry transfer), scanning electron microscopy (SEM), electron-beam evaporation (EBE) &amp; lithography (EBL), atomic force microscopy (AFM), sputtering, atomic layer etching (ALE), optical spectroscopy, soldering</p> <p><b>Programming:</b> MATLAB, Python, C++, HTML, CSS, JavaScript</p> <p><b>Other:</b> LaTeX, CAD (KLayout, AutoCAD), Microsoft Office Suite</p> <p><b>Languages:</b> English (native), Cantonese (native), Mandarin (fluent), German (elementary), Dutch (elementary)</p> |                   |
| PRESS<br>COVERAGE     | [2] <b>"CNAS visits Washington on a 'Call to Action!'"</b> ,<br><i>UC Riverside College of Natural &amp; Agricultural Sciences</i> , 2025.03                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |
|                       | [1] <b>"UCR team to participate in national undergraduate research program"</b> ,<br><i>by Iqbal Pittalwala, Inside UCR</i> , 2024.10                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |
| REFERENCES            | Available upon request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |